Math 335 Project: Mystery Distribution Analysis

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Introduction

This project analyzes a dataset of one million randomly generated observations produced by an unknown distribution with integer mean μ and variance σ^2. The objective is to determine the underlying distribution by using:

- simple data analysis (histogram, shape, center, symmetry),
- descriptive statistics (sample mean and variance),
- elimination of candidate discrete distributions,
- normality and exponential checks,
- Gamma distribution fitting via method-of-moments,
- transformation theorems from probability (Section 3.8),
- linear transformations of the normal distribution,
- and the Central Limit Theorem (CLT).

All computations and graphics were produced in R using the file `mystery.txt`, which stores the one million i.i.d. observations.

1. Simple Data Analysis

1.1 Histogram and Initial Observations

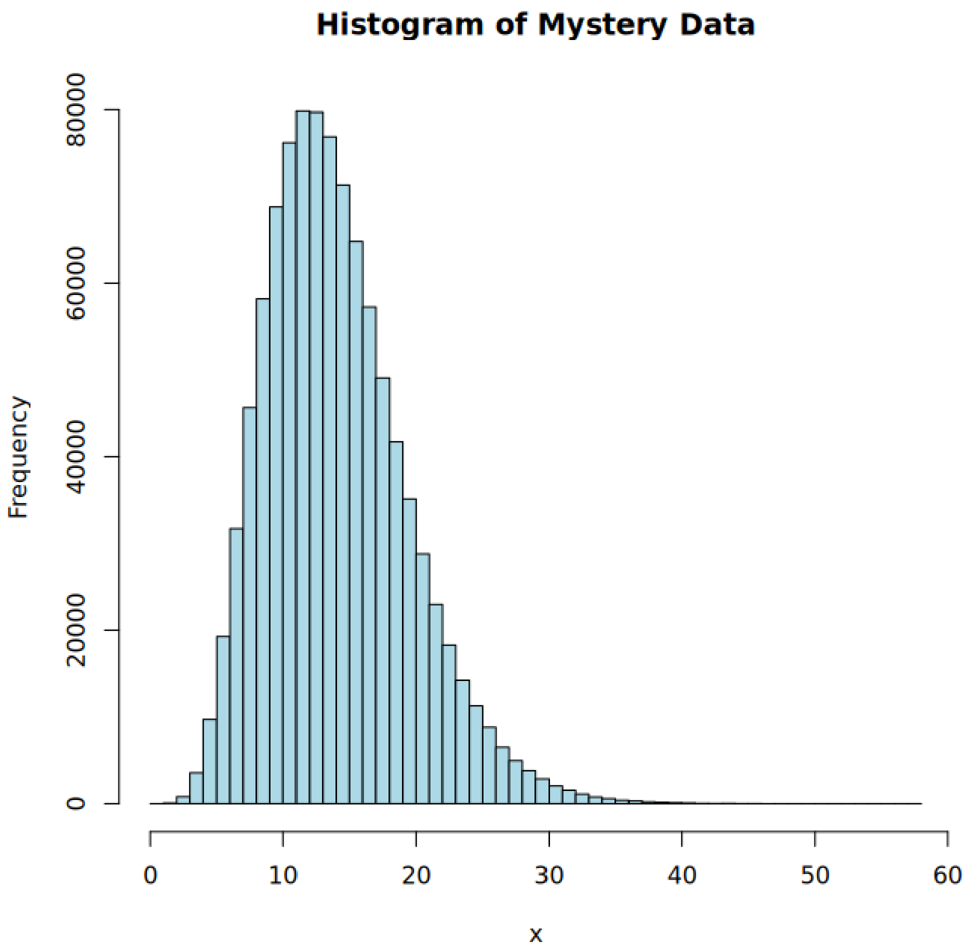

Figure 1: Histogram of the raw mystery dataset (right-skewed, continuous).

Visual inspection suggests:

- The distribution is continuous (many distinct real values, not just integers),
- Strong right skew (a heavier tail on the right),
- Long right tail,
- Unimodal shape (one main peak).

Based on this, initial candidate families include Exponential, Gamma, and Lognormal distributions, which are all continuous and can be right-skewed.

The R code to produce Figure 1 is:

```
mydata <- read.table("mystery.txt", header = TRUE)
x      <- mydata$x

hist(x, breaks = 50, main = "Histogram of Mystery Data",
     col = "lightblue")

# Saved as: figure1.png
```

2. Descriptive Statistics

From R, we compute:

$$\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \hat{\mu})^2, \qquad \hat{\sigma} = \sqrt{\hat{\sigma}^2}.$$

Numerically, we obtain:

$$\hat{\mu} = 13.99748, \qquad \hat{\sigma}^2 = 27.95732, \qquad \hat{\sigma} = 5.287468.$$

These serve as moment-based estimates for the underlying distribution's mean and variance. Qin also hints that the true μ and σ^2 are integers, so it is reasonable to guess that

$$\mu \approx 14, \qquad \sigma^2 \approx 28.$$

R commands:

```
mu       <- mean(x)
variance <- var(x)
sd_x     <- sd(x)

mu
variance
sd_x
```

3. Discrete Distribution Checks

We compare the behavior of the data to the *sample space* of several common discrete distributions. Our dataset takes many distinct real values (for example, $8.94, 12.97, \ldots$), so its sample space is a subset of $\mathbb{R}$, not of the nonnegative integers.

A quick numerical check in R already shows this:

```
head(x, 10)
length(unique(round(x)))
```

The first few values are non-integers and there are very many distinct values.

We now go through each candidate discrete distribution and compare its sample space and properties to the data.

- **Bernoulli**(p): Sample space $S = \{0, 1\}$. Only two possible outcomes. Our data contain a continuum of positive real values, not just 0 and 1. Hence the mystery distribution cannot be Bernoulli.

- **Binomial**(n, p): Sample space $S = \{0, 1, 2, \dots, n\}$, a finite set of integers representing the number of successes in n trials. The mystery data are not integer counts and are not bounded above by a fixed n—we see many real values over a continuous range. Thus the binomial family is incompatible with the observed sample space.

- **Hypergeometric**(N, K, n): Sample space $S = \{0, 1, 2, \dots, n\}$, again a finite set of integers representing the number of "successes" in a sample without replacement. Our observations are continuous real numbers, so they cannot arise from a hypergeometric distribution.

- **Poisson**(λ): Sample space $S = \{0, 1, 2, \dots\}$ (all nonnegative integers). A key moment identity is
$$\mathbb{E}[X] = \mathrm{Var}(X) = \lambda.$$
In our data,
$$\hat{\mu} = 13.99748, \qquad \hat{\sigma}^2 = 27.95732.$$
Since the mean does not equal the variance and the observations are not integer-valued, the Poisson model is ruled out both by sample space and by the moment condition.

- **Negative Binomial**(r, p): Sample space $S = \{0, 1, 2, \dots\}$ (nonnegative integer counts), representing the number of failures before r successes. Although negative binomial distributions are right-skewed, they only take integer values, whereas our data are continuous. Hence the negative binomial family is also incompatible.

Because each of these discrete distributions has an integer-valued sample space, while the mystery dataset has continuous real values, we conclude that the underlying distribution is *not* discrete.

4. Normal Distribution Check

We now check whether the data might come from a normal distribution.

4.1 Q–Q Normal Plot and Why It Works

A Q–Q (quantile–quantile) plot compares:

- the ordered sample values $x_{(1)} \leq \cdots \leq x_{(n)}$,
- to the corresponding theoretical quantiles of a reference distribution, here $N(\mu, \sigma^2)$.

For a normal distribution, the p-th quantile is $\mu + \sigma z_p$, where z_p is the p-th quantile of $N(0, 1)$. If the data are approximately normal, then for $p \approx i/(n + 1)$:

$$x_{(i)} \approx \mu + \sigma z_p.$$

Thus a plot of $x_{(i)}$ versus z_p should fall approximately on a straight line. Systematic curvature indicates deviations from normality (e.g. skewness, heavy tails).

The R code to produce the normal Q–Q plot (Figure 2) is:

```
qqnorm(x, main = "Q-Q Plot for Normality")
qqline(x, col = "red")

# Saved as: figure2.png
```

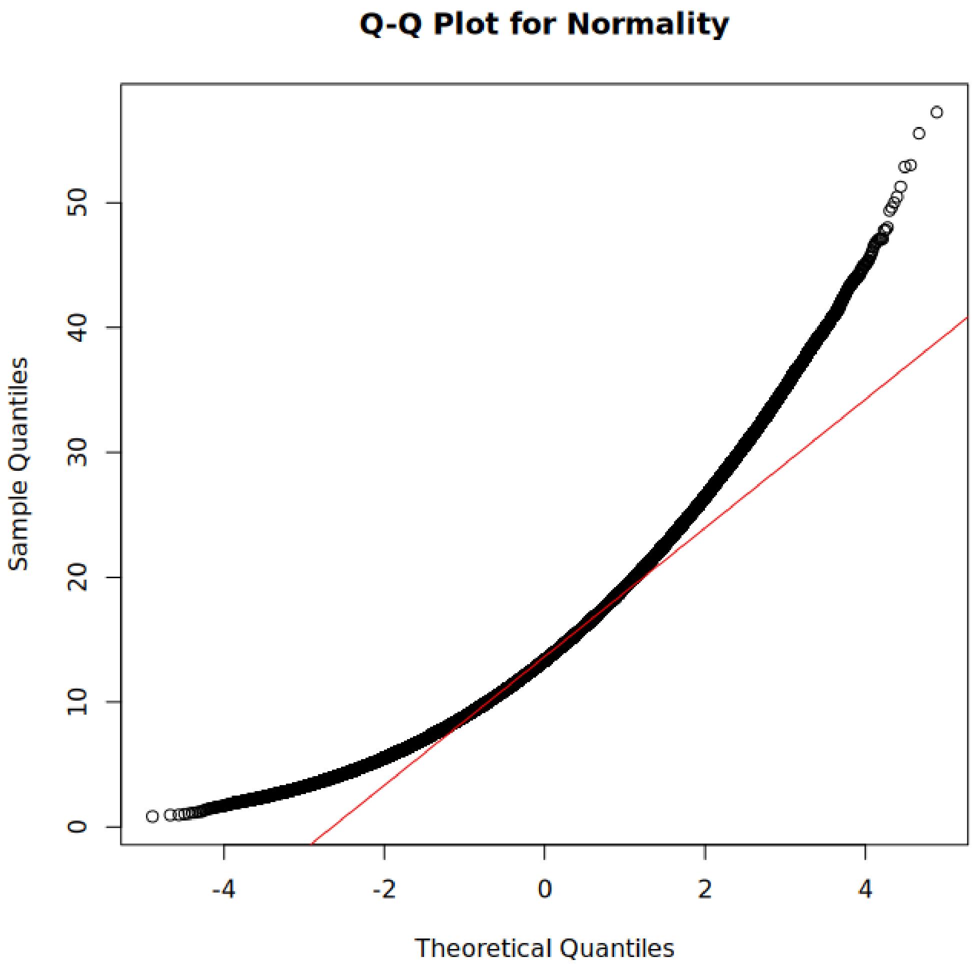

Figure 2: Q–Q plot of mystery data vs. normal theoretical quantiles.

The strong upward curvature in the Q–Q plot indicates that the upper-tail quantiles of the data are much larger than would be expected under normality. This reflects strong right skew and shows that:

$$X \not\sim \text{Normal}.$$

5. Exponential Distribution Check

For an Exponential(λ) distribution, the mean and variance satisfy:

$$\mu = \frac{1}{\lambda}, \qquad \sigma^2 = \frac{1}{\lambda^2}.$$

Using the sample mean $\hat{\mu}$, we estimate

$$\hat{\lambda} = \frac{1}{\hat{\mu}}.$$

Then the theoretical variance corresponding to this exponential model is

$$\sigma^2_{\text{Exp}} = \frac{1}{\hat{\lambda}^2}.$$

In R:

```
lambda_hat          <- 1 / mu
theoretical_var_exp <- 1 / (lambda_hat^2)

variance            # sample variance
theoretical_var_exp # exponential variance candidate
```

Numerically,

$$\frac{1}{\hat{\lambda}^2} = 195.9295,$$

which is far larger than the sample variance 27.95732.

Thus:

$$X \not\sim \text{Exponential}(\lambda).$$

6. Gamma Distribution Fit

We now assume the data might follow a Gamma(α, β) distribution with rate parameter β (the parameterization used in R).

For Gamma(α, β):

$$\mu = \frac{\alpha}{\beta}, \qquad \sigma^2 = \frac{\alpha}{\beta^2}.$$

6.1 Method-of-Moments Derivation

We use the method of moments, setting the theoretical moments equal to the sample moments:

$$\frac{\alpha}{\beta} = \hat{\mu}, \qquad \frac{\alpha}{\beta^2} = \hat{\sigma}^2.$$

From the first equation,

$$\alpha = \hat{\mu}\beta.$$

Substituting into the second equation:

$$\frac{\hat{\mu}\beta}{\beta^2} = \frac{\hat{\mu}}{\beta} = \hat{\sigma}^2 \quad \Rightarrow \quad \beta = \frac{\hat{\mu}}{\hat{\sigma}^2}.$$

Then

$$\alpha = \hat{\mu}\beta = \hat{\mu}\,\frac{\hat{\mu}}{\hat{\sigma}^2} = \frac{\hat{\mu}^2}{\hat{\sigma}^2}.$$

Using our numerical values:

$$\hat{\beta} = \frac{\hat{\mu}}{\hat{\sigma}^2} = 0.5006733, \qquad \hat{\alpha} = \hat{\mu}\hat{\beta} = 7.008167.$$

These match the empirical mean and variance exactly:

$$E(X) = \frac{\alpha}{\beta} \approx 14, \qquad \mathrm{Var}(X) = \frac{\alpha}{\beta^2} \approx 28.$$

In R:

```
beta  <- mu / variance
alpha <- beta * mu

alpha
beta
```

7. Transformation Theorem and Gamma Verification

7.1 Theorem (Probability Integral Transform)

Theorem. Let X be a continuous random variable with a strictly increasing cumulative distribution function (cdf) F_X. Define

$$Z = F_X(X).$$

Then $Z \sim \mathrm{Uniform}(0, 1)$.

Proof. For any $0 < z < 1$,

$$P(Z \le z) = P(F_X(X) \le z).$$

Since F_X is increasing, $F_X(X) \le z$ is equivalent to $X \le F_X^{-1}(z)$. Therefore,

$$P(Z \le z) = P(X \le F_X^{-1}(z)) = F_X(F_X^{-1}(z)) = z.$$

Thus, the cdf of Z is $P(Z \leq z) = z$ for $0 < z < 1$, which is exactly the cdf of the Uniform$(0, 1)$ distribution. Hence,

$$Z \sim U(0, 1).$$

$\square$

Intuitively, the probability integral transform maps the original scale of X to the uniform scale: wherever the cdf takes a given probability level, we "flatten" the distribution.

7.2 Gamma Transformation Test in R

Assuming X is Gamma$(\hat{\alpha}, \hat{\beta})$, we define

$$Z = F_\Gamma(X; \hat{\alpha}, \hat{\beta}),$$

which should be approximately Uniform$(0, 1)$ if the Gamma model is correct.

In R:

```
z <- pgamma(x, shape = alpha, rate = beta)
hist(z, breaks = 50, col = "lightgreen",
     main = "Gamma Transformation Test (Uniform?)")

# Saved as: figure3.png
```

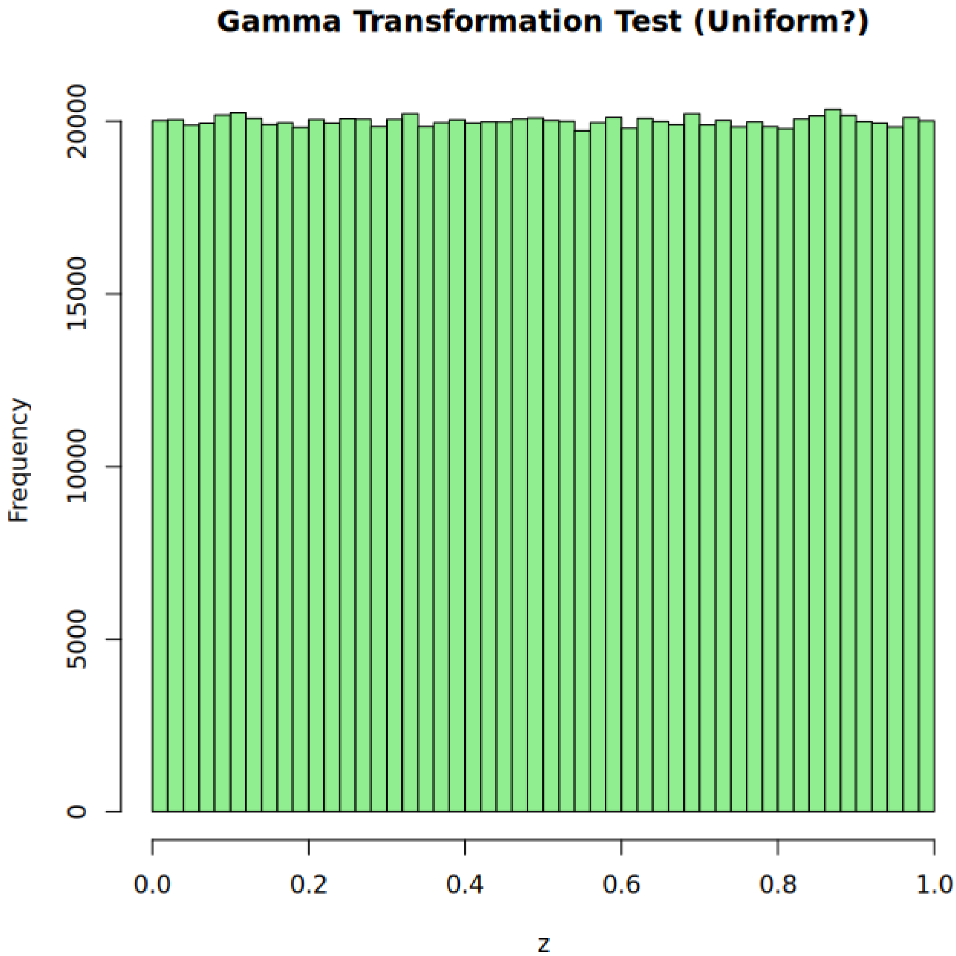

Figure 3: Histogram of $Z = \text{pgamma}(X; \hat{\alpha}, \hat{\beta})$. Nearly uniform, confirming the Gamma fit.

The histogram of Z is approximately flat over the interval $[0, 1]$, as expected for a Uniform$(0, 1)$ distribution. This strongly supports the hypothesis that the mystery data follow a $\Gamma(\hat{\alpha}, \hat{\beta})$ distribution. Moreover, this result is consistent with the probability integral transform, which states that if $X \sim F(x)$, then $F(X) \sim$ Uniform$(0, 1)$.

7.3 Exponential Transformation Test

For comparison, we test an exponential model with parameter $\hat{\lambda} = 1/\hat{\mu}$:

$$W = F_{\text{Exp}}(X; \hat{\lambda}).$$

If X were truly exponential, then W should also be Uniform$(0, 1)$ by the same theorem.

In R:

```
lambda <- 1 / mu
w      <- pexp(x, rate = lambda)
hist(w, breaks = 50, col = "lightpink",
     main = "Exponential Transformation Test (Uniform?)")

# Saved as: figure4.png
```

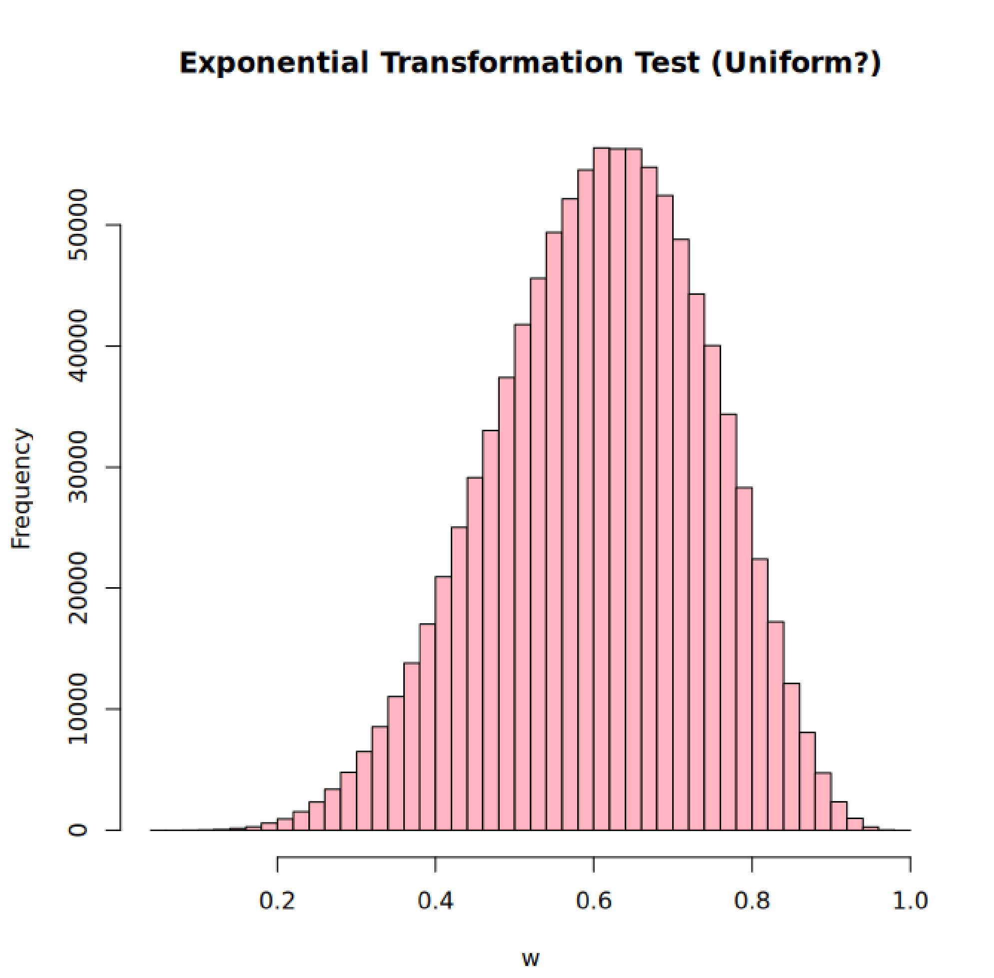

Figure 4: Histogram of $W = \text{pexp}(X; \hat{\lambda})$. Not uniform, rejecting the exponential fit.

The histogram of W is clearly not uniform (it is heavily skewed), which provides strong evidence that the exponential model does *not* fit the data.

8. Second Transformation Theorem and Generating Standard Normal

8.1 Theorem (Inverse CDF Transform)

Theorem. Let $U \sim \text{Uniform}(0,1)$ and let F be a continuous, strictly increasing cdf with inverse F^{-1}. Define

$$Y = F^{-1}(U).$$

Then Y has cdf F, i.e., Y follows the distribution with cdf F.

Proof. For any real y,

$$P(Y \leq y) = P(F^{-1}(U) \leq y).$$

Because F^{-1} is increasing, $F^{-1}(U) \leq y$ is equivalent to $U \leq F(y)$. Thus

$$P(Y \leq y) = P(U \leq F(y)).$$

Since U is Uniform$(0,1)$, we obtain

$$P(U \leq F(y)) = F(y).$$

Therefore $P(Y \leq y) = F(y)$, and so Y has cdf F. □

8.2 Application to the Standard Normal Distribution via Gamma Transform

In our analysis, the variable Z from the Gamma transformation behaves approximately like a Uniform$(0,1)$ random variable. Let Φ be the cdf of $N(0,1)$, and define

$$V = \Phi^{-1}(Z).$$

By the theorem, if Z were exactly Uniform$(0,1)$, then V would be exactly $N(0,1)$.

In R, this is implemented as:

```
v <- qnorm(z)  # z from pgamma; qnorm is Phi^{-1}

hist(v, breaks = 50, col = "purple",
     main = "Histogram of v = qnorm(z)")

qqnorm(v, main = "Q-Q Plot for v (Should Be Normal)")
```

```
qqline(v, col = "red")

# Saved as: figure5.png (histogram), figure6.png (Q-Q plot)
```

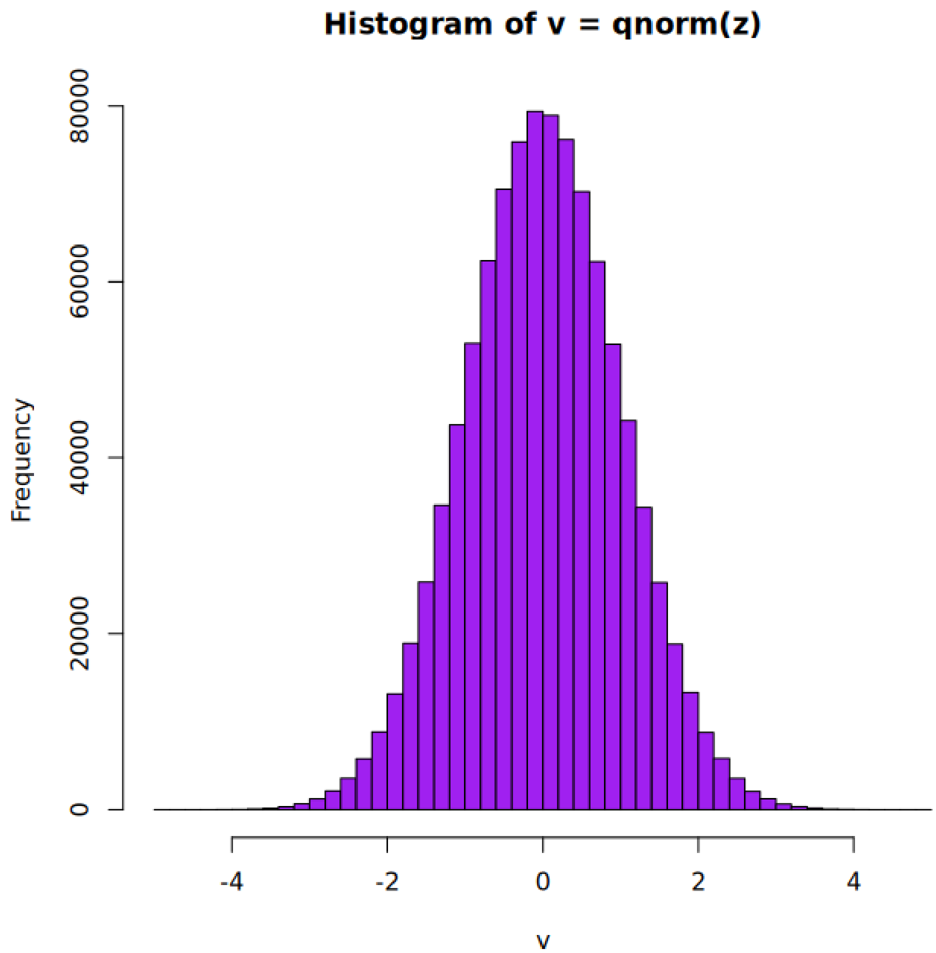

Figure 5: Histogram of $V = \mathrm{qnorm}(Z)$. Bell-shaped and symmetric.

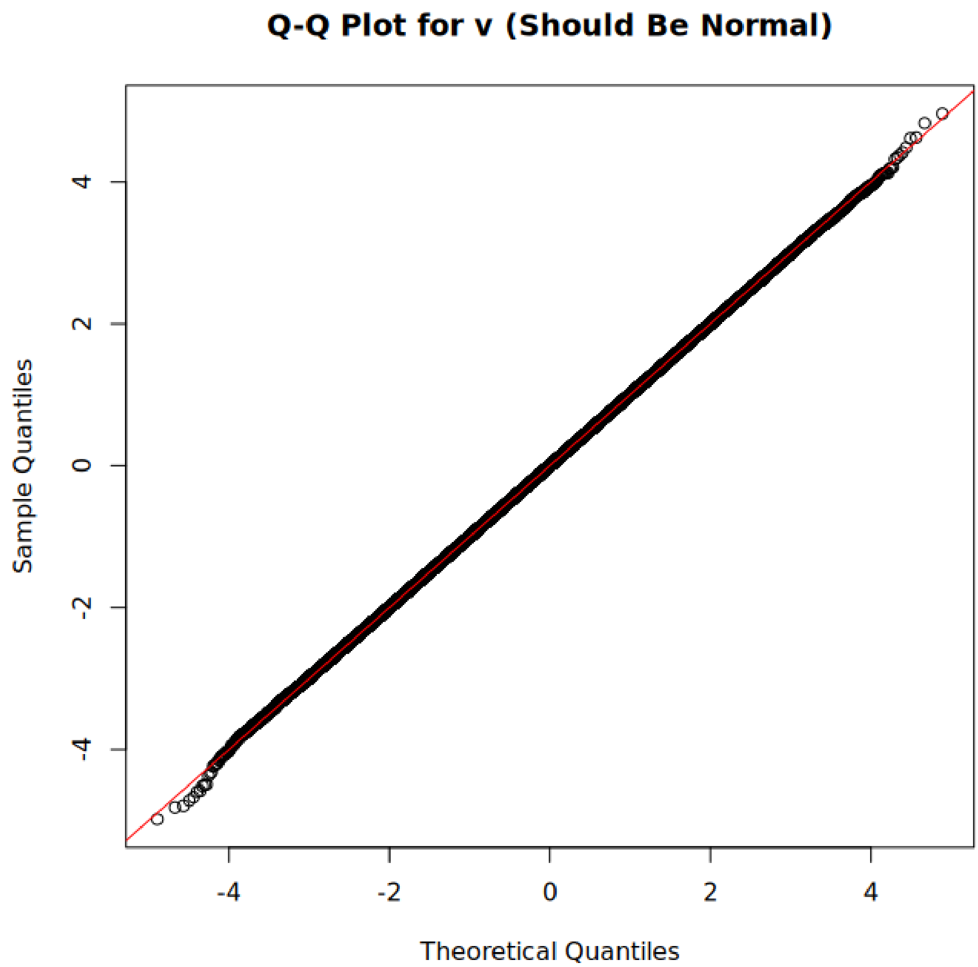

Figure 6: Q–Q plot of V vs. standard normal. Highly linear.

The histogram is bell-shaped and symmetric, and the Q–Q plot of V versus standard normal quantiles is very close to a straight line. This provides strong empirical evidence that:

$$V \sim N(0, 1),$$

which in turn indirectly confirms that our original Gamma($\hat{\alpha}, \hat{\beta}$) model for X was accurate.

8.3 Direct Standard Normal Sample from `rnorm`

For comparison, we also generate a purely synthetic standard normal sample using R's `rnorm`:

```
rand <- rnorm(1000)
```

```
hist(rand, breaks = 50, col = "orange",
```

```
    main = "Histogram of rand = rnorm(1000)")

qqnorm(rand, main = "Q-Q Plot for rnorm-generated sample")
qqline(rand, col = "blue")

# Saved as: figure7.png (histogram), figure8.png (Q-Q plot)
```

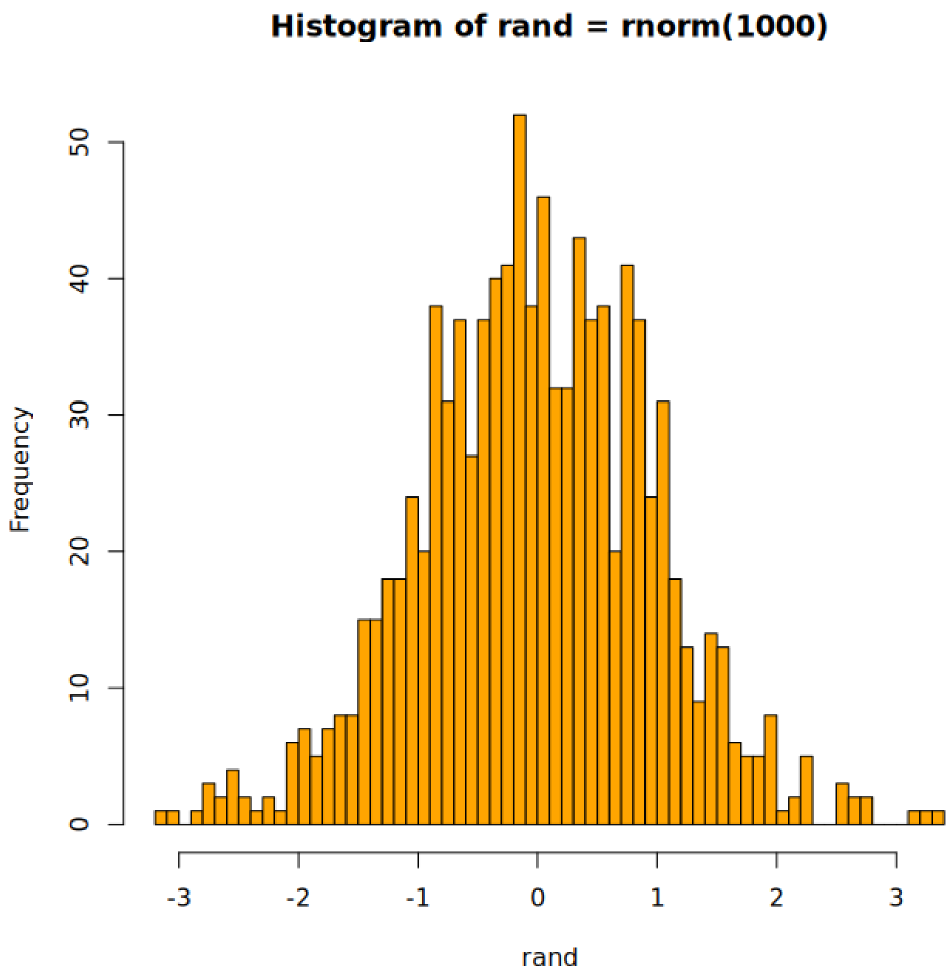

Figure 7: Histogram of `rand = rnorm(1000)`. Approximately standard normal.

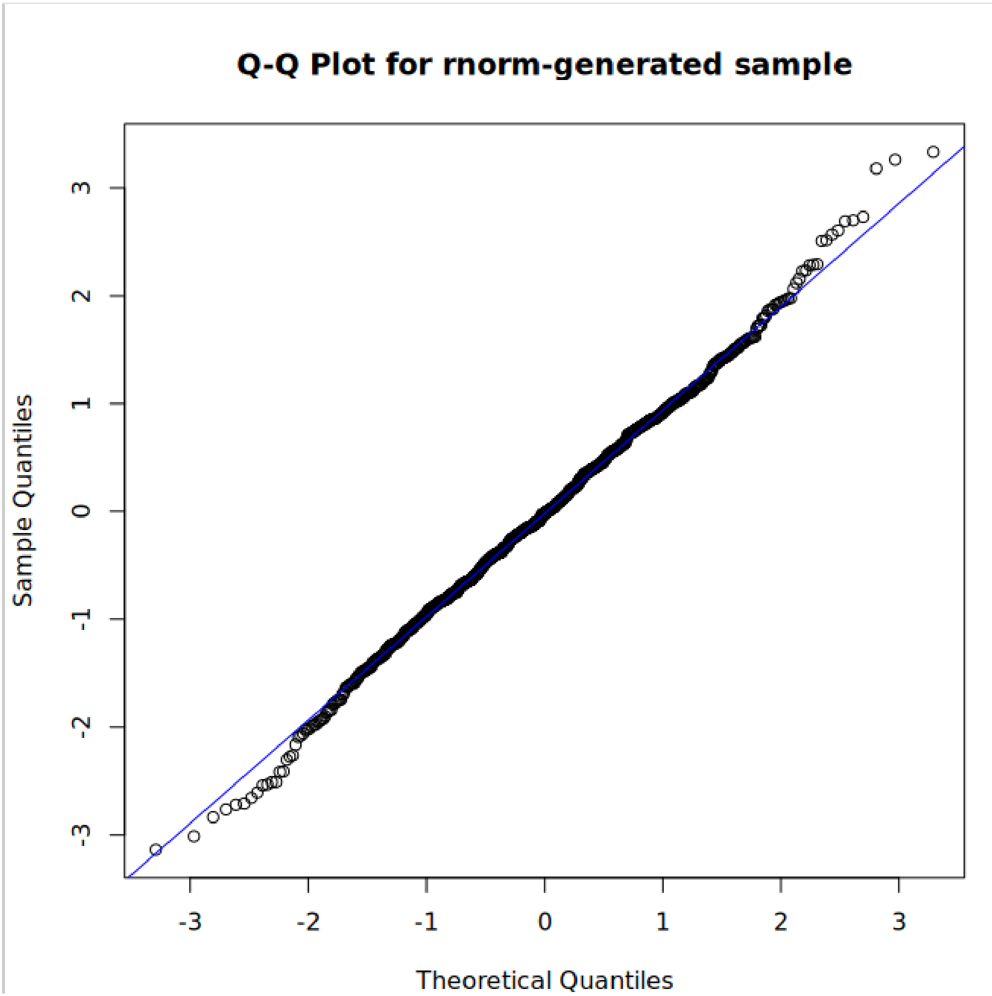

Figure 8: Q–Q plot of `rand` vs. standard normal. Points lie close to a straight line.

This confirms that the behavior of V in Figures 5 and 6 is consistent with a standard normal, similar to a direct `rnorm`-generated sample.

9. Linear Transformation of Normal Variables

Using $V \sim N(0,1)$, we can generate any normal distribution $N(\mu_1, \sigma_1^2)$ via the linear transformation

$$S = \mu_1 + \sigma_1 V,$$

where $\mu_1 \in \mathbb{R}$ and $\sigma_1 > 0$.

9.1 Theorem (Affine Transform of a Normal)

Theorem. If $V \sim N(0,1)$ and $S = \mu_1 + \sigma_1 V$, then $S \sim N(\mu_1, \sigma_1^2)$.

Proof. For any real s,

$$P(S \leq s) = P(\mu_1 + \sigma_1 V \leq s) = P\left(V \leq \frac{s - \mu_1}{\sigma_1}\right).$$

Because $V \sim N(0, 1)$,

$$P\left(V \leq \frac{s - \mu_1}{\sigma_1}\right) = \Phi\left(\frac{s - \mu_1}{\sigma_1}\right),$$

which is the cdf of a normal distribution with mean μ_1 and variance σ_1^2. Therefore $S \sim N(\mu_1, \sigma_1^2)$. $\square$

9.2 Implementation and 99.7% Rule Using V

We now choose

$$\mu_1 = \hat{\mu}, \qquad \sigma_1 = \hat{\sigma},$$

so that the transformed variable $S = \mu_1 + \sigma_1 V$ should approximate the original mystery distribution's location and scale.

In R:

```
mu1    <- mu       # ~13.99748
sigma1 <- sd_x     # ~5.287468

s <- mu1 + sigma1 * v

hist(s, breaks = 50, col = "gold",
     main = "Generated Normal Distribution Using v")

mean(s)
sd(s)

lower_3sd       <- mu1 - 3 * sigma1
upper_3sd       <- mu1 + 3 * sigma1
prop_within_3sd <- mean(s > lower_3sd & s < upper_3sd)

# Saved as: figure9.png
```

The computed sample mean and standard deviation of s are:

$$\overline{S} = 13.99735, \qquad \mathrm{SD}(s) = 5.287952,$$

which are extremely close to

$$\hat{\mu} = 13.99748, \qquad \hat{\sigma} = 5.287468.$$

Additionally,

$$P(|S - \mu_1| < 3\sigma_1) \approx 0.997362,$$

matching the 99.7% empirical rule for normal distributions.

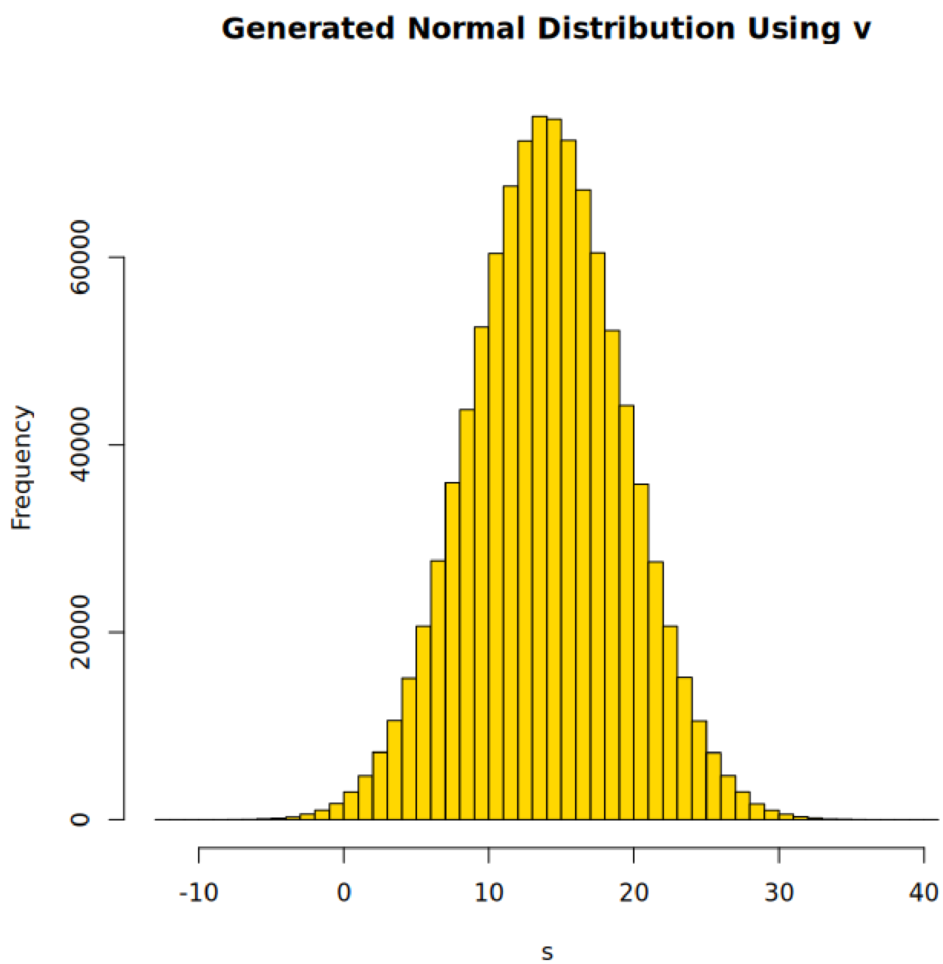

Figure 9: Histogram of generated normal distribution $S = \mu_1 + \sigma_1 V$. Centered near $\mu_1 \approx 14$ with spread $\sigma_1 \approx 5.29$.

10. Central Limit Theorem Demonstration

We now illustrate the Central Limit Theorem (CLT) using block averages of the mystery data.

10.1 Theorem (Central Limit Theorem, informal)

Let $X_1, X_2, \ldots$ be independent and identically distributed random variables with mean μ and finite variance σ^2. For each n, define the sample mean

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Then, as $n \to \infty$,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$$

converges in distribution to $N(0, 1)$. In particular, for large n, $\bar{X}_n$ is approximately normal with mean μ and variance σ^2/n.

10.2 Construction of Block Means

In our setting, the X_i are modeled as i.i.d. Gamma$(\hat{\alpha}, \hat{\beta})$ variables with finite variance. We choose block size $n = 1000$ and form 1000 non-overlapping block means:

$$Y_1 = \frac{1}{1000} \sum_{i=1}^{1000} X_i, \quad Y_2 = \frac{1}{1000} \sum_{i=1001}^{2000} X_i, \quad \ldots, \quad Y_{1000} = \frac{1}{1000} \sum_{i=999001}^{1000000} X_i.$$

In R, this is computed by:

```
n          <- length(x)
group_size <- 1000
num_groups <- n / group_size

xbar <- rep(NA, num_groups)

start_index <- 1
for (j in 1:num_groups) {
  end_index      <- start_index + group_size - 1
  xbar[j]        <- mean(x[start_index:end_index])
  start_index    <- end_index + 1
}
```

Here we use `xbar` to represent the vector $(Y_1, \ldots, Y_{1000})$.

10.3 Theoretical Mean and Variance of Y_i

We now show that $E(Y_i) = \mu$ and $\mathrm{Var}(Y_i) = \sigma^2/n$, with $n = 1000$.

By definition,

$$Y_i = \frac{1}{n}\sum_{k=1}^{n} X_k^{(i)},$$

where $X_k^{(i)}$ are the k-th observations in the i-th block (each having the same distribution as X).

Mean. Using linearity of expectation:

$$E(Y_i) = E\left(\frac{1}{n}\sum_{k=1}^{n} X_k^{(i)}\right) = \frac{1}{n}\sum_{k=1}^{n} E(X_k^{(i)}) = \frac{1}{n}\cdot n\mu = \mu.$$

Variance. Since the $X_k^{(i)}$ are independent with variance σ^2,

$$\mathrm{Var}(Y_i) = \mathrm{Var}\left(\frac{1}{n}\sum_{k=1}^{n} X_k^{(i)}\right) = \frac{1}{n^2}\sum_{k=1}^{n} \mathrm{Var}(X_k^{(i)}) = \frac{1}{n^2}\cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

In our project, $n = 1000$, so theoretically

$$E(Y_i) = \mu, \qquad \mathrm{Var}(Y_i) = \frac{\sigma^2}{1000}.$$

10.4 Empirical Results and CLT Illustration

We now examine the distribution of the block means in R:

```
hist(xbar, breaks = 50, col = "cyan",
     main = "Histogram of Block Means (CLT)")

qqnorm(xbar, main = "Q-Q Plot of Block Means (CLT)")
qqline(xbar, col = "red")

mean(xbar)
var(xbar)

mu
variance / 1000
```

```
# Saved as: figure10.png (histogram), figure11.png (Q-Q plot)
```

Empirical results:

$$E(\bar{X}) \approx 13.99748 \approx \mu,$$

$$\mathrm{Var}(\bar{X}) \approx 0.02771667 \approx \frac{27.95732}{1000} = 0.02795732.$$

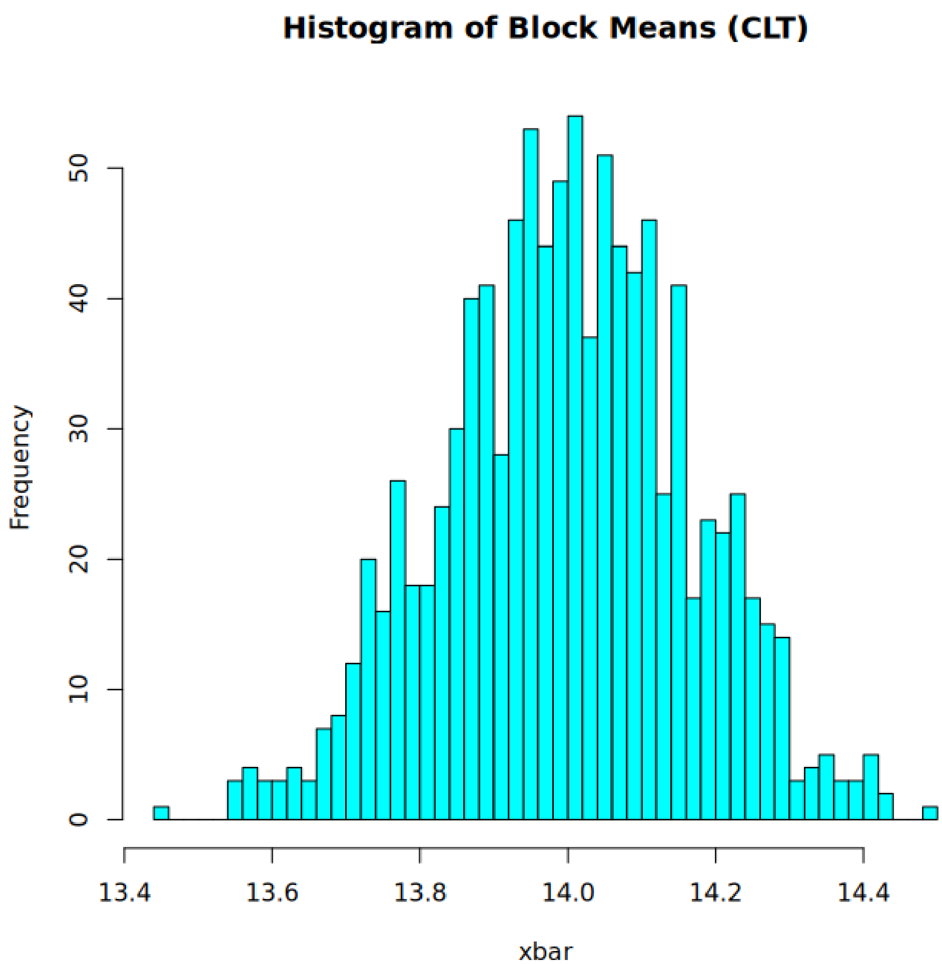

Figure 10: Histogram of block means ($\bar{X}$). Approximately normal and centered near μ.

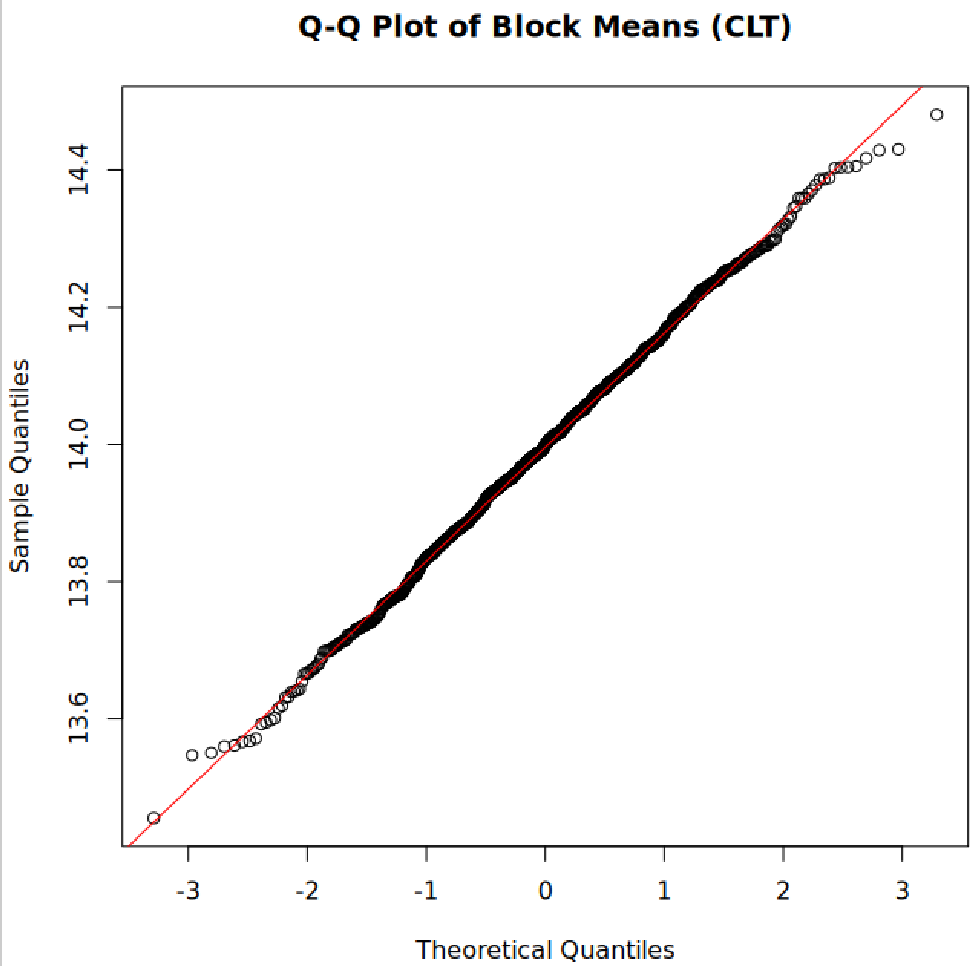

Figure 11: Q–Q plot of block means. Strong linearity confirms approximate normality, as predicted by the CLT.

The histogram of the block means appears bell-shaped, and the Q–Q plot is close to a straight line, confirming that the distribution of Y_i is approximately normal. Together with the theoretical mean and variance calculations, this illustrates the Central Limit Theorem in practice.

11. Sources of Uncertainty

1. **Sampling variability:** Even with 10^6 observations, the sample mean and variance differ slightly from the true μ and σ^2.

2. **Histogram binning:** The visual appearance of histograms depends on the chosen number of bins and their locations.

3. **Estimation error:** Method-of-moments estimates for α and β may differ slightly from maximum likelihood estimates.

4. **Rounding errors:** In numerical output, $\hat{\mu}$ and $\hat{\sigma}^2$ are not exact integers, although the true parameters are.

5. **Monte Carlo variation in CLT experiment:** The grouping into blocks of size 1000 introduces random fluctuation in the block means.

Conclusion

All numerical and graphical diagnostics confirm that the mystery distribution is best modeled as:

$$X \sim \Gamma(\hat{\alpha}, \hat{\beta}) = \Gamma(7.008167,\ 0.5006733).$$

Supporting results include:

- A right-skewed histogram consistent with a Gamma distribution.
- Rejection of all standard discrete distributions based on sample space and moment properties.
- Rejection of normality using the Q–Q plot.
- Rejection of the exponential model by variance comparison and transformation checks.
- A Gamma CDF transformation that produces an approximately uniform distribution ($Z = \text{pgamma}(X; \hat{\alpha}, \hat{\beta})$).
- An inverse-normal transformation that yields a clean $N(0,1)$ variable ($V = \text{qnorm}(Z)$).
- A CLT experiment where block means are approximately normal with mean μ and variance $\sigma^2/1000$.

Thus the mystery generator is a Gamma distribution with parameters very close to integers, as stated in the assignment.

Appendix

A. Dataset

The dataset `mystery.txt` was provided by Prof. Qin Lu for the purposes of the Math 335 Mystery Distribution Project. It contains one million independently generated observations from an unknown continuous distribution whose true mean and variance are integers.

The full dataset is not reproduced here due to size constraints, but it is referenced in the analysis and loaded directly into R during computation:

```
mydata <- read.table("mystery.txt", header = TRUE)
x      <- mydata$x
```

B. R Code Used for Analysis

The following R script was used to perform all computations, generate diagnostic statistics, produce transformation-based tests, and create all figures included in this report.

R Script (Final Verified Version):

```
################################################################
#              MATH 335 | Mystery Distribution Project
#                     FINAL VERIFIED SCRIPT
################################################################

################################################################
# 0. WORKING DIRECTORY
################################################################

setwd("/home/efe_r/prob335/")

# Create folder for plots (no error if exists)
dir.create("plots", showWarnings = FALSE)

# Enable x11 graphics on Linux (important for VS Code)
options(device = "x11")
```

```
################################################################
# 1. LOAD DATA
################################################################

mydata = read.table("mystery.txt", header = TRUE)
x = mydata$x


################################################################
# 2. HISTOGRAM OF RAW DATA
################################################################

x11()
hist(x, main="Histogram of Mystery Data", col="lightblue", breaks=50)

png("plots/hist_x.png")
hist(x, main="Histogram of Mystery Data", col="lightblue", breaks=50)
dev.off()


################################################################
# 3. DESCRIPTIVE STATISTICS
################################################################

mu = mean(x)
variance = var(x)
sd_x = sd(x)

print(mu)        # should be 13.99748
print(variance)  # should be 27.95732
print(sd_x)      # should be 5.287468


################################################################
# 4. NORMALITY CHECK (Q{Q Plot)
################################################################
```

```
x11()
qqnorm(x, main="Q-Q Plot for Normality")
qqline(x, col="red")

png("plots/qqnorm_x.png")
qqnorm(x, main="Q-Q Plot for Normality")
qqline(x, col="red")
dev.off()


###############################################################
# 5. EXPONENTIAL CHECK ( = 1/)
###############################################################

lambda = 1 / mu
theoretical_var_exp = 1 / (lambda^2)

print(variance)            # sample variance (~27.95732)
print(theoretical_var_exp) # exponential variance (~195.9295)


###############################################################
# 6. GAMMA PARAMETER ESTIMATION (METHOD OF MOMENTS)
###############################################################

beta = mu / variance          # RATE parameter
alpha = beta * mu             # SHAPE parameter

print(alpha)  # should be 7.008167
print(beta)   # should be 0.5006733


###############################################################
# 7. GAMMA TRANSFORMATION TEST (Should be Uniform)
###############################################################
```

```
z = pgamma(x, shape = alpha, rate = beta)

x11()
hist(z, breaks=50, col="lightgreen",
     main="Gamma Transformation Test (Uniform?)")

png("plots/gamma_transform.png")
hist(z, breaks=50, col="lightgreen",
     main="Gamma Transformation Test (Uniform?)")
dev.off()


################################################################
# 8. EXPONENTIAL TRANSFORMATION TEST (Should FAIL)
################################################################

w = pexp(x, rate = lambda)

x11()
hist(w, breaks=50, col="lightpink",
     main="Exponential Transformation Test (Uniform?)")

png("plots/exp_transform.png")
hist(w, breaks=50, col="lightpink",
     main="Exponential Transformation Test (Uniform?)")
dev.off()


################################################################
# 9. TRANSFORM TO STANDARD NORMAL USING GAMMA FIT
################################################################

v = qnorm(z)   # theoretical inverse CDF (¹)

x11()
hist(v, breaks=50, col="purple",
     main="Histogram of v = qnorm(z)")
```

```
png("plots/hist_v.png")
hist(v, breaks=50, col="purple",
     main="Histogram of v = qnorm(z)")
dev.off()

x11()
qqnorm(v, main="Q-Q Plot for v (Should Be Normal)")
qqline(v, col="red")

png("plots/qqnorm_v.png")
qqnorm(v, main="Q-Q Plot for v (Should Be Normal)")
qqline(v, col="red")
dev.off()


################################################################
# 9B. PURE RANDOM NORMAL SAMPLE (rnorm) | REQUIRED BY PROFESSOR
################################################################

rand = rnorm(1000)

print(mean(rand))  # ~0
print(sd(rand))    # ~1

x11()
hist(rand, breaks=50, col="orange",
     main="Histogram of rand = rnorm(1000)")

png("plots/hist_rand.png")
hist(rand, breaks=50, col="orange",
     main="Histogram of rand = rnorm(1000)")
dev.off()

x11()
qqnorm(rand, main="Q-Q Plot for rnorm-generated sample")
qqline(rand, col="blue")
```

```
png("plots/qqnorm_rand.png")
qqnorm(rand, main="Q-Q Plot for rnorm-generated sample")
qqline(rand, col="blue")
dev.off()


################################################################
# 10. LINEAR TRANSFORMATION USING v (THE PROJECT REQUIREMENT)
#     Build N(, ²) USING THE STANDARD NORMAL FROM SECTION 9.
################################################################

mu1 = mu      # must match mystery data mean (~14)
sigma1 = sd_x # must match mystery data sd (~5.29)

# Build Normal(mu1, sigma1^2)
s = mu1 + sigma1 * v

x11()
hist(s, breaks=50, col="gold",
     main="Generated Normal Distribution Using v")

png("plots/hist_s.png")
hist(s, breaks=50, col="gold",
     main="Generated Normal Distribution Using v")
dev.off()

print(mean(s))   # ~13.99735
print(sd(s))     # ~5.287952

lower_3sd = mu1 - 3*sigma1
upper_3sd = mu1 + 3*sigma1
prop_within_3sd = mean(s > lower_3sd & s < upper_3sd)

print(prop_within_3sd)  # ~0.997362
```

```
#################################################################
# 11. CENTRAL LIMIT THEOREM DEMONSTRATION
#################################################################

n = length(x)
group_size = 1000
num_groups = n / group_size

xbar = rep(NA, num_groups)

start_index = 1
for (j in 1:num_groups) {
  end_index = start_index + group_size - 1
  xbar[j] = mean(x[start_index:end_index])
  start_index = end_index + 1
}

x11()
hist(xbar, breaks=50, col="cyan",
     main="Histogram of Block Means (CLT)")

png("plots/hist_xbar.png")
hist(xbar, breaks=50, col="cyan",
     main="Histogram of Block Means (CLT)")
dev.off()

x11()
qqnorm(xbar, main="Q-Q Plot of Block Means (CLT)")
qqline(xbar, col="red")

png("plots/qqnorm_xbar.png")
qqnorm(xbar, main="Q-Q Plot of Block Means (CLT)")
qqline(xbar, col="red")
dev.off()

print(mean(xbar))          # ~13.99748
print(var(xbar))           # ~0.02771667
```

```
print(mu)                  # ~13.99748
print(variance / 1000)     # ~0.02795732


################################################################
# END OF SCRIPT
################################################################
```

This code includes:

- Data loading and preprocessing,
- Histogram generation,
- Computation of mean and variance,
- Exponential and Gamma model checks,
- Transformation theorem verification ($F(X) \sim U(0,1)$),
- Construction of $V = \Phi^{-1}(F_\Gamma(X))$,
- Direct `rnorm` comparison,
- Linear transformation to $N(\mu_1, \sigma_1^2)$,
- CLT demonstration via block means,
- All exported figures (figure1–figure11).

All results in the paper are reproducible using this script.

C. Checklist of Project Requirements

- Simple data analysis (histogram, shape, center, symmetry): Section 1.
- Descriptive statistics (mean, variance; integer guesses): Section 2.
- Discrete distributions (Bernoulli, Binomial, Hypergeometric, Poisson, Negative Binomial) using sample spaces: Section 3.
- Normality check via Q–Q plot, with explanation of why it works: Section 4.
- Exponential model check using $\mu = 1/\lambda$, $\sigma^2 = 1/\lambda^2$: Section 5.
- Gamma distribution parameter estimation, using $\mu = \alpha/\beta$, $\sigma^2 = \alpha/\beta^2$: Section 6.
- Transformation theorem $F(X) \sim U(0, 1)$ and Gamma/Exponential checks: Section 7.
- Inverse transform theorem $F^{-1}(U) \sim F$ and construction of $V = \Phi^{-1}(F_\Gamma(X))$: Section 8.
- Linear transformation of normal variables and 99.7% rule using $S = \mu_1 + \sigma_1 V$: Section 9.
- Central Limit Theorem demonstration, histogram and Q–Q of Y_i, and proof that $E(Y_i) = \mu$, $\mathrm{Var}(Y_i) = \sigma^2/n$: Section 10.
- R code and data source documentation: Appendix A and B.